INDUSTRIAL INTERNSHIP REPORT

Drugs, Side Effects, and Medical Condition Analysis

Project Domain:	Intern Role:	Technical Stack:
Data Science & Analytics	Business Analyst	Python, Scikit-Learn, SQL

1. Project Objective

The primary goal of this project was to analyze the complex interplay between pharmaceutical drugs, their documented side effects, and the medical conditions they are prescribed for. By utilizing data from ***drugs.com***, the study sought to uncover patterns in drug efficacy (ratings), safety profiles (pregnancy categories), and market availability (Rx vs. OTC).

2. Data Methodology

The analysis followed a structured pipeline starting from raw data ingestion to predictive modeling. Key technical steps included:

- **Data Imputation:** Handled over 7,400 missing values, particularly in brand names and alcohol interaction columns.
- **Normalization:** Converted activity percentages into a decimal scale $0.0 \leq x \leq 1.0$ for consistent model input.
- **Statistical Analysis:** Validated distribution shifts in ratings across different drug classes.

3. Key Findings

ID	Insight Description
1	**Pain, Colds & Flu, and Acne** represent the conditions with the highest drug availability in the dataset.
2	Drug ratings are **right-skewed**; the majority of medications maintain a user rating between 6 and 8.

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3	**Prescription drugs (Rx)** dominate the market landscape, accounting for approximately 75% of the data.
4	**Hives and difficult breathing** emerged as the most frequently reported adverse side effects across categories.
5	A high prevalence of **Pregnancy Category C and D** drugs indicates significant fetal risk across common treatments.
6	The **Random Forest model** yielded superior accuracy in classifying high-performing drugs compared to linear models.
7	Feature importance analysis identified **Review Count** and **Activity Level** as the strongest predictors of drug rating.
8	**K-Means Clustering** successfully segmented the inventory into 3 distinct groups based on user engagement and efficacy.
9	Drugs with **alcohol interactions** demonstrated unique rating distributions, suggesting impact on patient satisfaction.
10	**Upper respiratory combinations** and **topical acne agents** are the most frequent drug classes identified.

4. Future Roadmap & Recommendations

Strategic Next Steps

- **NLP Integration:** Implement TF-IDF or Word2Vec on the ***side_effects*** corpus to extract nuanced safety signals.
- **Recommendation Engine:** Build a collaborative filtering system to suggest drugs based on condition-safety compatibility.
- **Deployment:** Transition these insights into a real-time **Streamlit or Flask** dashboard for clinical decision support.
- **Association Rules:** Utilize the Apriori algorithm to discover hidden correlations between co-occurring side effects.